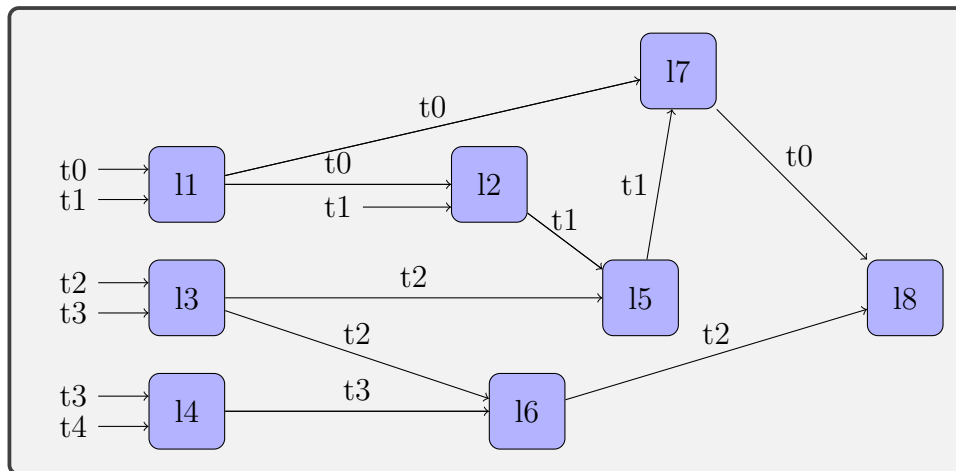


Exercise Sheet

Task 1 – Data dependency graph

- a) Create a data dependency graph as seen in the lecture for the following instruction sequence:

I1: ADD t0, t0, t1
I2: SLL t1, t0, t1
I3: SUB t2, t2, t3
I4: ADD t3, t3, t4
I5: MUL t1, t1, t2
I6: XOR t2, t2, t3
I7: AND t0, t0, t1
I8: ADD t0, t0, t2



- b) How many execution phases are needed at least?
For this task, you may assume that there are always enough free functional units.

The longest path in the graph is I1, I2, I5, I7, I8 and 5 instructions long, therefore 5 execution cycles are needed to run the sequence.

Task 2 – Dependencies

For an instruction sequence:

I1: ADD t0, t0, t1
I2: MUL t1, t1, t2
I3: SRL t3, t0, t3
I4: AND t2, t2, t4
I5: XOR t1, t4, t5
I6: SUB t5, t5, t6
I7: ADD t4, t4, t5
I8: SLL t6, t3, t6

Decide for each of the following re-ordered sequences if it can be made valid with register renaming. If that is not the case, explain why. If it is possible, rename all necessary registers with virtual registers v0-vn.

	I1	I2	I3	I4	I5	I6	I7	I8
I1		A(t1)	T(t0)		A(t1)			
I2				A(t2)	O(t1), A(t1)			
I3								T(t3)
I4							A(t4)	
I5						A(t5)	A(t4)	
I6							T(t5)	A(t6)
I7								
I8								

a) I1, I3, I8, I4, I5, I7, I6, I2

This sequence cannot be executed as the true dependence between I6 and I7 on register t5 is violated

b) I1, I3, I8, I6, I7, I2, I4, I5

This sequence can be executed with the following renamings:
Anti-Dependence (I4, I7): Let I7 write to v0
Anti-Dependence (I5, I6): Let I6 write to v1 and I7 read from v1
Anti-Dependence (I5, I7): Let I7 write to v0
Anti-Dependence (I6, I8): Let I8 write to v2

c) I2, I4, I1, I6, I7, I8, I5, I3

This sequence cannot be executed as the true dependence between I3 and I8 on register t3 is violated